FYODOR MALKOV

PostgreSQL Database Administrator | DBA / Infrastructure Engineer

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PostgreSQL • High Availability • Disaster Recovery • ClickHouse • Ansible • Linux • Database Reliability • Security

PROFESSIONAL SUMMARY

Database Administrator and Infrastructure Engineer with 13+ years in IT and hands-on experience operating high-load production systems and terabyte-scale databases. **Core specialization: PostgreSQL** and its surrounding infrastructure — replication, high availability, backup and recovery, monitoring, security, and automation. Previously supported a large enterprise estate of 600+ database instances, including HA PostgreSQL environments based on Patroni, Consul/etcd, and HAProxy. Currently modernizing PostgreSQL and ClickHouse production infrastructure in a regulated critical information infrastructure (CII) environment serving mobile network operators, with a focus on fault tolerance, disaster recovery, security, operational reliability, and Ansible-based automation.

PROFESSIONAL EXPERIENCE

FSII National Research Centre of Telecommunication (NRCT) | Database Administrator

Aug 2026 – Present

Administer and modernize PostgreSQL and ClickHouse production infrastructure supporting critical telecommunications data in a regulated CII environment.

- Administer terabyte-scale PostgreSQL production infrastructure: replication, monitoring, backup and recovery, troubleshooting, and operational support.
- Maintain ClickHouse production infrastructure, including replication, ClickHouse Keeper, backups, monitoring, and cluster health.
- Analyze and modernize a legacy PostgreSQL architecture to improve fault tolerance, manageability, and readiness for future load growth.
- Design the target PostgreSQL high-availability and automated failover architecture using Patroni and distributed configuration storage; evaluate supporting load-balancing components.
- Develop and improve backup and disaster-recovery processes with Barman; restored a standby after an incident involving unavailable WAL and returned replication to a healthy state.
- Develop monitoring for PostgreSQL, ClickHouse, and infrastructure components in Zabbix, including replication health, workload, and operational metrics.
- Build Ansible automation for deployment and configuration of database, backup, monitoring, security, and HA-related components.
- Participate in database security hardening and in defining operational procedures, recovery scenarios, and technical documentation for CII systems.

Open-source contribution: implemented secure Mamonsu-to-Zabbix metric delivery using TLS PSK and certificates; submitted upstream as postgrespro/mamonsu PR #235.

Rostelecom | Database Administrator

Dec 2023 – Feb 2026

Administered enterprise database infrastructure for a large telecommunications company, supporting 600+ database instances across production, pre-production, development, and test environments.

- Supported PostgreSQL-compatible platforms, Postgres Pro, Pangolin, ClickHouse, and Cassandra, from small instances to multi-terabyte production systems.
- Administered standalone and highly available PostgreSQL environments based on Patroni, Consul/etcd, HAProxy, and load-balancing infrastructure.
- Performed backup and recovery operations using pg_probackup; investigated incidents and resolved operational database issues.
- Configured and maintained monitoring with Zabbix and Tantor Platform, including agent deployment, upgrades, metric analysis, and alert investigation.
- Developed Python and Ansible automation for DBA operations and large-scale administrative tasks; adapted workflows for corporate AWX.
- Performed database security hardening: reduced superuser access, implemented RBAC, enabled TLS, and integrated access with Active Directory.
- Worked with NetBox, Vault, Jira, OTRS, and Confluence in enterprise operational workflows.

Scientific Centre for Family Health and Human Reproduction Problems | IT Infrastructure & Systems Administration Specialist (Part-Time)

Sep 2020 – Jan 2026

- Administered Linux virtual machines and Proxmox VE; participated in migration from VMware to an in-house Proxmox environment.
- Managed backups and supported PostgreSQL-based services including Moodle, GitLab, internal Django applications, and BigBlueButton; maintained REDCap on MySQL.
- Supported research, healthcare, and educational IT infrastructure and provided technical consultation to users.

Horizon LLC | Systems Engineer — Laboratory Information Systems

Jun 2022 – Oct 2023

- Administered Windows/Linux servers and supported PostgreSQL, Oracle, and MongoDB through administrative SQL, troubleshooting, log analysis, backups, and monitoring.
- Automated routine tasks with Python and integrated laboratory analyzers and electronic-signature workflows.

EARLIER EXPERIENCE

Irkutsk National Research Technical University — Lecturer | Sep 2020 – Jun 2021 — Delivered undergraduate laboratory classes and supported practical technical training.

Institute of System Dynamics and Control Theory, SB RAS — Programmer | Jun 2018 – Dec 2018 — Developed Django applications, administered Redmine and GitLab, designed databases, and prepared software-registration documentation.

Irkutsk National Research Technical University — Junior Researcher | Feb 2017 – Aug 2021 — Research and software-development projects, publications and grant work; exposure to OpenStack and storage systems.

Irkutsk Scientific Centre, SB RAS — Junior Researcher | Jun 2015 – Dec 2017 — Worked on grant-funded research and software projects, requirements documentation, and scientific publications.

UMNIK Grant Project — Software Developer | Mar 2015 – May 2015 — C++/Qt development, database design, OpenCV-based image analysis, and grant reporting.

Irkutsk National Research Technical University — Research Laboratory Assistant / Technician Programmer | May 2011 – Apr 2013 — Administered local networks, workstations, servers, computer labs, and laboratory equipment.

SELECTED OPEN-SOURCE & ENGINEERING WORK

Independent engineering work complements core DBA experience with systems programming, networking, automation, and AI-assisted software engineering. **Selected areas:** Python and Rust tooling, rule-based network routing R&D, GitHub Actions / CI/CD, architecture and data-model design, and controlled use of OpenAI Codex and Claude Code for requirements analysis, implementation, code review, debugging, testing, and documentation.

TECHNICAL SKILLS

Databases: PostgreSQL, Postgres Pro, Pangolin, ClickHouse, Cassandra; basic hands-on experience with Oracle, MongoDB, MySQL

High Availability: Patroni, Consul, etcd, HAProxy, PostgreSQL replication, ClickHouse replication, ClickHouse Keeper

Backup & Recovery: Barman, pg_probackup, PostgreSQL backup/recovery, standby recovery, disaster-recovery procedures

Monitoring: Zabbix, Mamonsu, Tantor Platform

Automation & Infrastructure: Ansible, AWX, Python, Bash, Linux, Docker, Proxmox, Git, GitHub Actions, Vault, NetBox

Security: TLS, RBAC, Active Directory integration, database hardening, CII security requirements

EDUCATION, TRAINING & LANGUAGES

Irkutsk National Research Technical University

Information Systems and Technologies — 2017

Automated Information Processing and Control Systems — 2015

Professional training: ClickHouse Development and Support (2025); Cassandra Fundamentals / EAS-014 (2024)

Languages

Russian — Native

English — B2; technical reading/writing stronger than speaking/listening